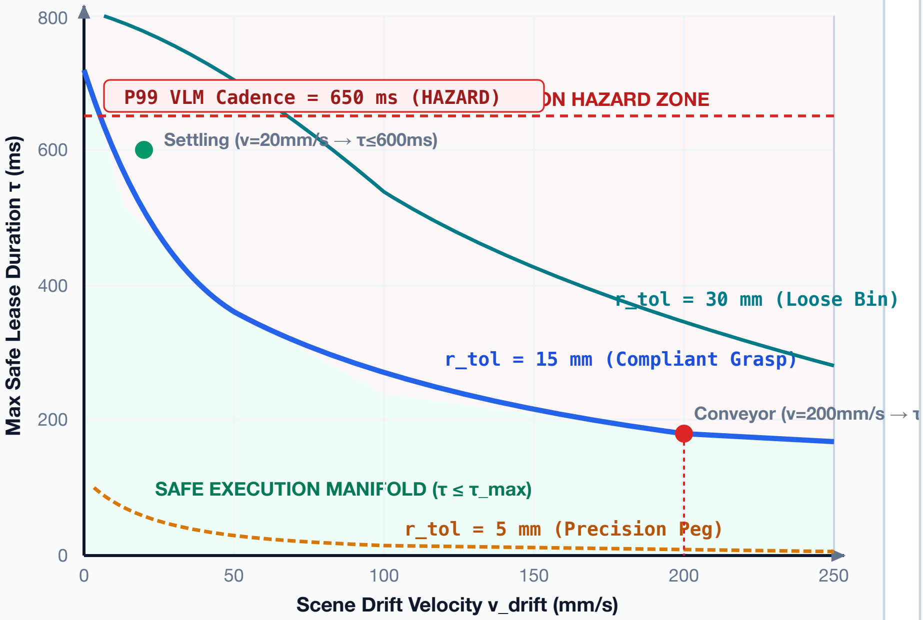
INTENT LEASE DYNAMICS & EARLY ADMISSIBILITY FILTERING

Physics-Derived TTL Horizon vs. O(1) Kinematic & Dynamic Ingestion Gateways

SCENE PHYSICS · DYNAMICS

PANEL A: LEASE DURATION vs. DRIFT



Mathematical Derivation: Maximum Safe TTL (§10.4)

$\tau_{max} = (r_{tol} - \sigma_{sensor}) / v_{drift}$

Design Rule: Lease validity is an invariant of **scene physics**, not model compute speed.
If inference latency jitters (P99 > τ_{max}), system must halt smoothly rather than stretch lease.

INGESTION GATE · ARCHITECTURE

PANEL B: EARLY REJECTION FILTER

SYSTEM 2 VLM / BRAIN — 168 Bytes →

`Intent Packet (seq, t_src, T∈SE(3), Σ, τ, F_max)`

< 1 μs — 1. Monotonicity & Hash Gate → Drop Stale/Replay

seq > seq_curr ∧ parent_hash == ring_buffer[t_src]

O(1) · 5 μs — 2. Kinematic Workspace & Occupancy → Out-of-Reach Reject

$p_{target} \in \mathcal{W}_{reach}$ ∧ Occupancy(p_{target}) == VALID

< 2 μs — 3. Dynamic Reachability Bound → Time Insufficient

$t_{min} = 2\sqrt{(d/a_{max})} \leq \tau_{rem}$ (Under thermal limits)

< 1 μs — 4. Covariance & Ambiguity Gate → Refusal on Distractor

$\lambda_{max}(\Sigma) \leq \sigma^2_{max}$ ∧ Softmax Entropy $H(Y) \leq H_{thresh}$

✓ ADMITTED INTENT RECORD

- Dispatched to 50 Hz Trajectory Optimizer
- Enforced by 1000 Hz Real-Time MCU Shield

`Total Ingestion Overhead: 42 μs (P99)`

✗ STRUCTURED REJECTION

- Diagnostic Feedback to System 2 Reasoner
- Returns Shortfall (t_{min} - τ) & Margin

`Prevents Solver Queue Starvation`

Computational Cost Asymmetry (§10.5)

Early Rejection Gate (Ingestion): `42 μs` scalar processor lookup. Zero queue bubble.

Late Rejection (Planner Interior): `15–80 ms` solver penalty search. Starves real-time pipeline.

Ingestion filtering guarantees downstream solver only executes kinematically solvable goals.